# Expert Analysis: Journal White Paper on Entanglement Entropy Density as a Gravitational Source

Tasks 1–5: Framework Stress Test, Theoretical Tightening, OP3 Project Plan, Experimental Sharpening, and Publication Strategy

Expert Physics Review

Responding to: K. Monette, *Entanglement Entropy Density as a Gravitational Source*, Draft March 19, 2026

March 2026

## Contents

**Task 1: Framework Restatement and Stress Test** **2**

**1 Task 2: Theoretical Tightening** **2**
1.1 Consistency check of the covariant formulation . . . . . 2
1.2 The critical tension in the effective action . . . . . 3
1.3 Proposed improvement to the effective action . . . . . 4
1.4 Why a potential $V(\phi)$ and curvature coupling are not needed . . . . . 5

**2 Task 3: OP3 — Full Lindblad Project Plan** **5**
2.1 Platform choice and justification . . . . . 5
2.2 Precise definitions for the OP3 program . . . . . 5
2.3 Staged numerical research plan . . . . . 6
2.4 How OP3 outputs plug back into the white paper . . . . . 9

**3 Task 4: Experimental Proposal Sharpening** **9**
3.1 Fully specified experimental configuration . . . . . 9
3.2 Timing and observable . . . . . 10
3.3 Systematic error budget: fatal vs. non-fatal . . . . . 11

**4 Task 5: Publication Strategy and Gap Analysis** **11**
4.1 Paper I: Theory Letter (4–6 pages) . . . . . 11
4.2 Paper II: Technical Backup (20–30 pages) . . . . . 12
4.3 Paper III: Experimental Proposal (12–15 pages) . . . . . 12
4.4 Timeline . . . . . 13

**Summary** **13**

# Task 1: Framework Restatement and Stress Test

The white paper introduces a phenomenological extension of general relativity in which the internal entanglement structure of a laboratory quantum device contributes to the gravitational stress-energy tensor. The coupling is mediated by a coarse-grained scalar field $\phi(x) \equiv S_{\rm ent}(x)$ representing the density of entanglement between subsystems of the device (bits per cubic meter), treated as an emergent, effective degree of freedom rather than a new fundamental field. This quantity is operationally defined: it is the von Neumann entropy $S_{\rm int}(t; A) = -k_{\rm B}\,{\rm Tr}_A[\rho_A \ln \rho_A]$ of a chosen bipartition $A|B$ of the device, divided by $k_{\rm B} \ln 2$ and by the cell volume. Crucially, the paper correctly identifies that only candidate (3) — the internal inter-subsystem entanglement entropy — has the right behavior: it *decreases* under decoherence (as $\rho_D$ approaches a product state over subsystems), unlike total von Neumann entropy or system–environment entropy, which both increase.

The coupling to gravity is introduced through a minimal effective action containing a kinetic term for $\phi$ and a linear potential $\alpha\phi$, ensuring that variation with respect to $g^{\mu\nu}$ produces a stress-energy tensor of the cosmological-constant type: $T^{\rm ent}_{\mu\nu} = \Lambda_{\rm ent}(x) g_{\mu\nu}$ with $\Lambda_{\rm ent} = \tilde{\kappa} c^4 S_{\rm ent}/(8\pi G k_{\rm B} \ln 2)$. The paper uses the action to argue that total stress-energy is covariantly conserved: the scalar field equation of motion $\beta\Box\phi = \alpha$ ensures $\nabla^\mu T^{\rm total}_{\mu\nu} = 0$ on-shell. The Bianchi identity then implies that ordinary matter is *not* independently conserved where $\nabla S_{\rm ent} \neq 0$ — the framework predicts energy-momentum exchange between matter and the entanglement sector. The paper correctly notes that the specific estimate $\tilde{\kappa} = -1/4$ is a motivated ansatz under a Planck-scale holographic regulator hypothesis and is not derivable from first principles; $\tilde{\kappa}$ is treated as a free phenomenological parameter throughout.

For the radial structure, the paper correctly shows that the entanglement-induced acceleration correction scales as $\Delta a(R) \propto M_{\rm ent}(<R)/R^2$ for any bounded source, resolving the earlier confusion about $1/R$ vs. $1/R^2$. The experimental program is appropriately modest: NISQ-scale devices with 50 qubits are the near-term target; macroscopic BEC interferometry is demoted due to extreme spatial screening. The falsification criterion is reformulated as a bound on the observable combination $|\tilde{\kappa}_{\rm eff}|\,\Delta M_{\rm ent}/R^2_{\rm eff}$ rather than a direct bound on $\tilde{\kappa}$, which is the correct framing given that $\alpha_{\rm screen}$ (the ratio of actual to ideal time-integrated entanglement) is currently unknown.

**What the framework does not claim:** It does not derive $\tilde{\kappa}$ from first principles. It does not assert that the scalar field $\phi$ is fundamental. It does not claim that existing equivalence-principle experiments (MICROSCOPE, Asenbaum et al.) directly constrain $\tilde{\kappa}$, since their test masses have negligible and nearly identical $S_{\rm ent}$. It does not provide the Lindblad derivation of $\alpha_{\rm screen}$ (this is explicitly flagged as an open problem). The paper is admirable in these omissions.

# 1 Task 2: Theoretical Tightening

## 1.1 Consistency check of the covariant formulation

The stress-energy definition and conservation statements are consistent with standard GR conventions. The decomposition $T^{\rm total}_{\mu\nu} = T^{\rm matter}_{\mu\nu} + T^{\rm ent}_{\mu\nu}$ is well-posed. The metric signature is

not stated in the white paper; this should be fixed as $(-,+,+,+)$ in the final manuscript to remove ambiguity for readers coming from different conventions. Specifying $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ explicitly in the definition of the modified field equation is also advisable.

Three implicit assumptions about $S_{\text{ent}}(x)$ need to be stated explicitly:

1. **Smoothness.** The coarse-graining procedure in Section 2.3 produces a piecewise-constant field on cells of volume $\Delta V$. The paper's Poisson equation analysis assumes $\rho_{\text{ent}}(x)$ is smooth. A sentence should state: "We treat $S_{\text{ent}}(x)$ as smooth at scales large compared to the device lattice spacing; this is the effective-field-theory approximation appropriate below the Planck scale."

2. **Bipartition dependence.** The definition $S_{\text{int}}(t; A)$ depends on the choice of bipartition $A|B$. The paper does not specify how this choice is made across spacetime or whether a canonical choice exists. For a uniform linear chain of qubits, the natural choice is the half-chain bipartition; for a 2D array, a geometric half-plane. This must be stated explicitly, otherwise $S_{\text{ent}}(x)$ is not well-defined as a scalar field.

3. **Locality.** The coarse-grained field $S_{\text{ent}}(x)$ is assigned to a spatial cell, but entanglement entropy is a nonlocal quantity (it depends on correlations across the cut, not just within the cell). The paper should note that the coarse-graining breaks translational invariance in a hardware-dependent way and that $S_{\text{ent}}(x)$ is therefore not a local observable in the usual QFT sense — it is a *collective, coarse-grained* quantity.

## 1.2 The critical tension in the effective action

**Expert Flag**

The paper's effective action has a fundamental tension that must be addressed before publication.

The scalar field $\phi(x) \equiv S_{\text{ent}}(x)$ is simultaneously:

(a) **Externally prescribed:** $\phi(x)$ is determined by the quantum state of the device, prepared and controlled by experimenters. It is *not* a freely propagating degree of freedom.

(b) **Dynamical:** The action's variation with respect to $\phi$ yields the field equation $\beta\Box\phi = \alpha$, which is an autonomous dynamics for $\phi$ independent of the quantum state.

These are incompatible. If $\phi$ is externally prescribed (as it must be for experimental control), then varying the action with respect to $\phi$ is not a valid operation — the field equation is not satisfied by the experimentally prepared state.

Specifically: for a cluster state of 50 qubits, $S_{\text{ent}}(x)$ is a specific function of space determined by the quantum circuit. This function does not satisfy $\beta\Box S_{\text{ent}} = \alpha$ in general. The action's $\phi$-variation is therefore fictitious.

**Two resolutions exist:**

**Resolution A (Constrained action).** Treat $\phi$ as a *fixed external source* (not dynamical), and add it to the action only through the coupling $\alpha\phi\sqrt{-g}$. Then the field equation is

dropped, and $T^{\text{ent}}_{\mu\nu}$ is computed from the variation with respect to $g^{\mu\nu}$ only, giving simply $T^{\text{ent}}_{\mu\nu} = \alpha\phi g_{\mu\nu} - \frac{\beta}{2}(\nabla\phi)^2 g_{\mu\nu} + \beta\nabla_\mu\phi\nabla_\nu\phi$. Conservation is ensured not by the $\phi$ equation of motion but by the identity $\nabla^\mu G_{\mu\nu} = 0$ together with the specific $\phi$ profile determined by the quantum state.

**Resolution B (Source term).** Promote $\phi$ to a fully dynamical field sourced by the quantum device: add a source term $J(x)\phi$ to the action, where $J(x)$ encodes how the quantum device drives the entanglement sector. Then $\phi$ satisfies $\beta\Box\phi = \alpha + J(x)$, and the quantum state enters through $J(x)$. In this case, $\phi$ deviates from $S^{\text{quantum}}_{\text{ent}}$ by the propagator of the Klein-Gordon equation — an interesting physical prediction, but also a significant complication.

**Recommended:** Resolution A. It preserves the original physical picture (experimentally controlled $\phi$), avoids the fictitious field equation, and keeps the framework minimal. The kinetic term should be retained because it contributes to the stress-energy and gives $T^{\text{ent}}_{\mu\nu}$ a physically richer structure than pure $\Lambda_{\text{ent}} g_{\mu\nu}$.

## 1.3 Proposed improvement to the effective action

**Proposed Improvement**

**Replacement text for Section 3.3 of the white paper.**

To provide an action-based derivation of $T^{\text{ent}}_{\mu\nu}$ without treating $\phi$ as a free dynamical field, we adopt the following approach. We introduce $\phi(x) = S_{\text{ent}}(x)$ as an *external scalar source field* determined by the quantum state of the device — it is not varied in the action. The effective action for the gravitational sector, in the presence of the fixed source $\phi(x)$, is:

$$S[g;\phi] = \int d^4x\sqrt{-g}\left[\frac{c^3}{16\pi G}R - \frac{\beta}{2}(\nabla\phi)^2 - \alpha\phi\right], \tag{1}$$

where $\alpha \equiv \tilde{\kappa}c^4/(8\pi G k_{\text{B}} \ln 2)$ and $\beta > 0$ is a coupling with dimensions of (energy density) $\times$ (length/entropy)$^2$. Varying with respect to $g^{\mu\nu}$ alone gives the Einstein equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left[T^{\text{matter}}_{\mu\nu} + \beta\big(\nabla_\mu\phi\nabla_\nu\phi - \tfrac{1}{2}g_{\mu\nu}(\nabla\phi)^2\big) + \alpha\phi\, g_{\mu\nu}\right]. \tag{2}$$

The entanglement stress-energy is:

$$T^{\text{ent}}_{\mu\nu} = \beta\big(\nabla_\mu\phi\nabla_\nu\phi - \tfrac{1}{2}g_{\mu\nu}(\nabla\phi)^2\big) + \alpha\phi\, g_{\mu\nu}. \tag{3}$$

Conservation of total stress-energy, $\nabla^\mu T^{\text{total}}_{\mu\nu} = 0$, is guaranteed by $\nabla^\mu G_{\mu\nu} \equiv 0$ and does not require any field equation for $\phi$. Instead, it imposes a consistency condition:

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\nabla^\mu T^{\text{ent}}_{\mu\nu} = -\beta\,(\Box\phi)\,\nabla_\nu\phi + \beta\,\nabla_\nu\big(\tfrac{1}{2}(\nabla\phi)^2\big) - \alpha\,\nabla_\nu\phi. \tag{4}$$

The right-hand side is a specific function of the externally prescribed $\phi(x)$ profile. It predicts an anomalous force on matter wherever $\nabla S_{\text{ent}} \neq 0$, with magnitude proportional to $\alpha\,|\nabla S_{\text{ent}}|$. This is a concrete, composition-independent fifth-force prediction that can be bounded experimentally.

In the limit $\nabla\phi \approx 0$ (slowly varying entanglement density), Eq. (3) reduces to $T^{\text{ent}}_{\mu\nu} \approx \alpha\phi\, g_{\mu\nu} = \Lambda_{\text{ent}} g_{\mu\nu}$, recovering the cosmological-constant ansatz as the leading approximation. The kinetic term $\beta(\nabla\phi)^2$ provides the next correction, physically encoding the energy cost of entanglement-density gradients.

### 1.4 Why a potential $V(\phi)$ and curvature coupling are not needed

The reviewer prompt asks whether a potential $V(\phi)$ or curvature coupling $\xi R\phi^2$ should be added. The answer is: not in the first paper. Both would introduce additional free parameters without adding falsifiable content at NISQ scales. A curvature coupling $\xi R\phi$ (linear in $\phi$) would be degenerate with the $\alpha\phi$ term in Eq. (1) at leading order in perturbation theory. A nonlinear potential $V(\phi) = m^2\phi^2/2$ would give $\phi$ a mass $m$, causing the entanglement field to be Yukawa-suppressed at distances $R \gg \hbar/(mc)$. Unless there is a physical reason to expect such a mass, it should not be introduced. These are appropriate additions for the technical backup paper (Paper II) as model extensions to be constrained by experiment.

## 2 Task 3: OP3 — Full Lindblad Project Plan

### 2.1 Platform choice and justification

We choose a **fixed-frequency superconducting transmon chain** (IBM Quantum architecture) for the following reasons:

- Hamiltonian and Lindblad parameters are publicly available from hardware calibration data
- Gate set tomography (GST) data exists for characterizing Kraus maps
- Tensor network methods (MPO-TDVP) are well-developed for 1D chains up to $N \sim 100$
- Circuit depth can be varied continuously and systematically

### 2.2 Precise definitions for the OP3 program

**System Hamiltonian** (in rotating frame, two-level approximation):

$$H_{\text{sys}} = \sum_{k=1}^{N} \frac{\omega_k}{2}\sigma_k^z + \sum_{k=1}^{N-1} J_k\big(\sigma_k^+\sigma_{k+1}^- + \text{h.c.}\big)\,, \tag{5}$$

with $\omega_k/2\pi \in [4.5, 5.0]\,\text{GHz}$ (sampled from published IBM calibration) and $J_k/2\pi = 5\,\text{MHz}$ (nearest-neighbor exchange coupling via bus resonator).

**Lindblad operators and rates:**

$$L_{1,k} = \sqrt{\gamma_1}\,\sigma_k^-, \qquad \gamma_1 = 1/T_1 \approx 2\pi \times 5\,\text{kHz} \quad (T_1 = 200\,\mu\text{s}), \tag{6}$$

$$L_{\phi,k} = \sqrt{\gamma_\phi/2}\,\sigma_k^z, \qquad \gamma_\phi = 1/T_\phi \approx 2\pi \times 8\,\text{kHz} \quad (T_\phi = 125\,\mu\text{s}). \tag{7}$$

These are state-of-the-art parameters from IBM Falcon-R10 and Eagle processors (2024–2025).

**Master equation:**

$$\dot{\rho} = -\frac{i}{\hbar}[H_{\text{sys}}, \rho] + \sum_{k=1}^{N} \sum_{\alpha \in \{1,\phi\}} \left( L_{\alpha,k} \rho L_{\alpha,k}^{\dagger} - \tfrac{1}{2}\{L_{\alpha,k}^{\dagger} L_{\alpha,k}, \rho\} \right). \tag{8}$$

**Definition of $M_{\text{ent}}(t)$ for this platform:**

For a chain of $N$ qubits, we use the half-chain bipartition $A = \{1, \ldots, N/2\}$, $B = \{N/2+1, \ldots, N\}$:

$$M_{\text{ent}}(t) = S_{\text{int}}(t; A) = -k_{\text{B}} \operatorname{Tr}_A[\rho_A(t) \ln \rho_A(t)], \quad \rho_A(t) = \operatorname{Tr}_B \rho(t). \tag{9}$$

**Screening factor:**

$$\alpha_{\text{screen}}(N, \tau) = \frac{\int_0^{\tau} M_{\text{ent}}(t)\, dt}{\int_0^{\tau} M_{\text{ent}}^{\text{ideal}}(t)\, dt}, \tag{10}$$

where $M_{\text{ent}}^{\text{ideal}}(t)$ is computed with $\gamma_1 = \gamma_\phi = 0$ (ideal unitary evolution).

**Initial states for the program:**

- **Cluster state (1D):** $|\text{Cl}\rangle = \prod_{\langle jk \rangle} \text{CZ}_{jk} |+\rangle^{\otimes N}$. Ideal half-chain entropy: $M_{\text{ent}}^{\text{ideal}} = k_{\text{B}} \ln 2$ (maximal for any bipartition of a 1D cluster state).
- **GHZ state:** $|\text{GHZ}\rangle = (|0\rangle^{\otimes N} + |1\rangle^{\otimes N})/\sqrt{2}$. Ideal half-chain entropy: $M_{\text{ent}}^{\text{ideal}} = k_{\text{B}} \ln 2$ (same magnitude, but different decay behavior).
- **Product state (Branch B):** $|+\rangle^{\otimes N}$. Ideal entropy: $M_{\text{ent}}^{\text{ideal}} = 0$.

## 2.3 Staged numerical research plan

**Open Problem**

**Stage 1 (Months 1–2): $N \leq 10$, exact diagonalization.**
**Method:** Full Lindblad master equation (8) evolved via matrix exponential (Expokit or SciPy `scipy.linalg.expm`) on the $4^N$-dimensional Liouvillian. For $N = 10$: Liouvillian dimension $= 4^{10} \approx 10^6$; feasible on a single workstation.
**Computations:**

1. For initial states $|\text{Cl}\rangle$ and $|\text{GHZ}\rangle$ on $N \in \{2, 4, 6, 8, 10\}$ qubits:
   (a) Evolve $\rho(t)$ under (8) for $0 \leq t \leq 5 \max(T_1, T_\phi)$.
   (b) Compute $M_{\text{ent}}(t)$ via (9) at each time step.
   (c) Compute $\alpha_{\text{screen}}(N, \tau)$ via (10) for $\tau = T_\phi, 2T_\phi, 5T_\phi$.
2. Also compute the *site-resolved* entropy $s_k(t) = -k_{\text{B}} \operatorname{Tr}_k[\rho_k(t) \ln \rho_k(t)]$ to understand spatial structure of entanglement decay.

**Output:** Plots of $M_{\text{ent}}(t)$ and $\alpha_{\text{screen}}(N, \tau)$ vs. $N$ and $\tau$ for both cluster and GHZ states. Comparison of decay rates. Identification of which state gives larger $\alpha_{\text{screen}}$ at fixed $N, \tau$.

**Expected finding:** Cluster states should show $\alpha_{\text{screen}} \sim e^{-\gamma_\phi \tau}$ (single-qubit decay time $T_\phi$) while GHZ states show $\alpha_{\text{screen}} \sim e^{-N\gamma_\phi \tau}$ (collective decay time $T_\phi/N$). Quantitative verification of this prediction is the primary output of Stage 1.

**Open Problem**

**Stage 2 (Months 2–4): $N \leq 20$, gate-error Kraus maps.**
**Method:** Replace ideal unitary gates in the cluster state preparation circuit with their Kraus-map representations, obtained from gate set tomography (GST) data. GST data is publicly available for IBM devices from published calibration reports.
**Additional modeling:** For each two-qubit CZ gate on qubits $(j, j+1)$, replace $U_{\text{CZ}}$ with a Kraus map $\mathcal{E}_{\text{CZ}}^{(j)}$ characterized by a $4 \times 4$ process matrix $\chi^{(j)}$. The gate error rate $\varepsilon_{2Q} \approx 5 \times 10^{-3}$ (IBM Falcon, 2025) enters as:

$$\mathcal{E}_{\text{CZ}}(\rho) = (1 - \varepsilon_{2Q}) U_{\text{CZ}} \rho U_{\text{CZ}}^\dagger + \varepsilon_{2Q}\, \mathcal{D}_{\text{depol}}(\rho), \tag{11}$$

where $\mathcal{D}_{\text{depol}}$ is a local depolarizing map.
**Computations:**

1. For $N \in \{10, 15, 20\}$ qubits, prepare the cluster state circuit using Kraus maps.
2. Evolve under (8) with the gate errors applied at each gate layer.
3. Compute $M_{\text{ent}}(t)$, $\alpha_{\text{screen}}(N, \tau)$, and the *total entanglement budget*:

$$\mathcal{E}_{\text{total}}(N, d) \equiv \int_0^\tau \sum_k s_k(t)\, dt, \tag{12}$$

   where $d$ is the circuit depth. Optimize $d$ to maximize $\mathcal{E}_{\text{total}}$.

**Output:** $\alpha_{\text{screen}}(N, \tau)$ as a function of $N$, $\tau$, and circuit depth $d$. A fit of the form $\alpha_{\text{screen}} \approx A\, e^{-N^\alpha \gamma_\phi \tau - d\varepsilon_{2Q}}$ to identify scaling exponents $\alpha$ and the relative contributions of decoherence vs. gate errors. The value of $\alpha$ determines whether $\alpha_{\text{screen}}$ scales polynomially (cluster state, $\alpha \approx 0$) or exponentially (GHZ, $\alpha = 1$) in $N$.

**Open Problem**

**Stage 3 (Months 4–8): $N = 50$, tensor network methods.**
**Method:** For $N = 50$, the Liouvillian Hilbert space dimension is $4^{50} \approx 10^{30}$, far beyond exact diagonalization. Use matrix product operator (MPO) methods:

1. Represent $\rho(t)$ as an MPO with bond dimension $\chi$.
2. Evolve under the Lindblad equation using the time-dependent variational principle (TDVP) for MPOs (implemented in TeNPy, ITensor, or quimb).
3. Monitor entanglement entropy of $\rho(t)$ and $\rho_A(t)$ at each time step.

**Bond dimension requirements:** For cluster states under 1D Lindblad dynamics, the operator entanglement entropy of $\rho(t)$ grows logarithmically in time, suggesting that MPO bond dimension $\chi \sim 10^2$–$10^3$ should suffice for $N = 50$ up to $t \sim T_\phi$. For GHZ states, the rapid entanglement collapse makes the dynamics easier (lower bond dimension after decoherence begins).
**Computations:**

1. Prepare the $N = 50$ cluster state as an MPS with bond dimension $\chi_0 = 2$ (exact for the cluster state).
2. Convert to MPO representation for the Lindblad evolution.
3. Evolve for $0 \leq t \leq 3T_\phi$; compute $M_{\text{ent}}(t)$ and $\alpha_{\text{screen}}(50, \tau)$.
4. Repeat for product state (Branch B) as control.

**Output:** Full $\alpha_{\text{screen}}(N, \tau)$ curve for $N = 50$. Extrapolation to the $N \to \infty$ limit if the scaling law from Stage 2 holds. A definitive answer to the question: is $\alpha_{\text{screen}}$ nonzero and significant for a 50-qubit cluster state on a transmon platform with realistic noise?
**The pivotal result from Stage 3 is:** If $\alpha_{\text{screen}}(50, T_\phi) \gtrsim 10^{-2}$, the experimental program is viable. If $\alpha_{\text{screen}} \lesssim 10^{-4}$, the NISQ experiment will not produce a useful bound and the framework's near-term experimental program must be reconsidered.

**Open Problem**

**Stage 4 (Months 8–12): Analytic bounds and scaling theory.**
Using the results of Stages 1–3, derive or conjecture:

1. A *lower bound* on $\alpha_{\text{screen}}$ for cluster states using the quantum data processing inequality: $S_{\text{int}}(t; A) \geq S_{\text{int}}(0; A) - \gamma_\phi \, \xi \, t$ for some correlation length $\xi$, where the inequality follows from the fact that local dephasing can only reduce entanglement at a rate bounded by the dephasing rate times the correlation length.
2. A *critical* $N$ at which $\alpha_{\text{screen}}$ drops below 1% of its ideal value. From the Stage 2 scaling law $\alpha_{\text{screen}} \approx A \, e^{-\gamma_\phi \tau}$ for cluster states: $N^* \sim |\ln(0.01)|/\gamma_\phi \tau =$ a computable number.
3. The *optimal circuit*: the cluster state preparation circuit $C^*$ that maximizes $\mathcal{E}_{\text{total}}(N, d)$ for fixed $N$ and hardware parameters. This is an optimization problem over the space of native-gate sequences.

**Output:** A table of $(N, d^*, \alpha^*_{\text{screen}})$ for the optimal cluster state circuits on the IBM transmon platform. A scaling law $\alpha_{\text{screen}}(N) \approx f(N)$ with analytic form. Plots showing the tradeoff between $N$ (larger $M_{\text{ent}}$) and $\alpha_{\text{screen}}$ (smaller due to decoherence), with the optimal operating point $N^*$.

### 2.4 How OP3 outputs plug back into the white paper

Once $\alpha_{\text{screen}}(N, \tau)$ is computed, the following replacements can be made in the white paper:

- Section 5.1: Replace "$\alpha_{\text{screen}}$ is an open parameter" with "For a 50-qubit transmon cluster state with $T_1 = 200\,\mu$s, $T_2 = 100\,\mu$s, and $\tau = T_\phi/2$, we find $\alpha_{\text{screen}} = [numericalvalue] \pm [error]$ from MPO-Lindblad simulation."
- Section 6: Replace the bound on $|\tilde{\kappa}_{\text{eff}}|\,\Delta M_{\text{ent}}/R_{\text{eff}}^2$ with an explicit numerical value using the computed $\alpha_{\text{screen}}$ and measured $\Delta M_{\text{ent}}$.
- Paper III (experimental proposal): Include the $\alpha_{\text{screen}}$ calculation as a theoretical prediction that the experiment should test. A significant deviation of measured $\alpha_{\text{screen}}$ from the Lindblad prediction would itself be a result — it would indicate either new physics or an error in the open-system model.

# 3 Task 4: Experimental Proposal Sharpening

### 3.1 Fully specified experimental configuration

**Platform:** $6 \times 6 = 36$ qubit $^{87}$Rb neutral-atom array in reconfigurable optical tweezers. Lattice spacing: $a = 5\,\mu$m. Two-qubit gates via Rydberg blockade ($|70S_{1/2}\rangle$, blockade radius $r_b = 8\,\mu$m for nearest-neighbor coupling only).

**Qubit basis:** $|0\rangle = |5S_{1/2}, F = 1, m_F = 0\rangle$, $|1\rangle = |5S_{1/2}, F = 2, m_F = 0\rangle$ (clock transition, $\omega_{\text{hf}}/2\pi = 6.835\,\text{GHz}$).

**Branch A (high entanglement): 2D cluster state.**

1. Initialize: $|+\rangle^{\otimes 36}$ via $\pi/2$ pulses on clock transition.
2. Apply CZ gates to all nearest-neighbor horizontal pairs (one layer, $\tau_{\text{CZ}} = 200\,\text{ns}$ per gate, parallelized: total time $200\,\text{ns}$).
3. Apply CZ gates to all nearest-neighbor vertical pairs (one layer, $200\,\text{ns}$).
4. Result: 2D cluster state. Half-array entanglement entropy $M_{\text{ent}}^A \approx 6k_{\text{B}} \ln 2$ (for a $3 \times 6$ vs $3 \times 6$ bipartition).
5. Total preparation time: $\tau_{\text{prep}} \approx 1\,\mu$s.

**Branch B (low entanglement): Matched product state.**

1. Initialize: $|+\rangle^{\otimes 36}$ (same as Branch A).
2. Apply CZ gates to all horizontal pairs with *zero interaction time* (Rydberg laser on but detuned by $\Delta_{\text{off}} = 10\Omega$, giving gate fidelity $\approx (1 - \Omega^2/\Delta_{\text{off}}^2) = 99\%$ for the identity operation).
3. Apply the same "null" vertical CZ gates.

4. Result: Product state $|+\rangle^{\otimes 36}$ with $M_{\text{ent}}^B = 0$.

**Why this Branch A/B pair:** The 2D cluster state and the $|+\rangle^{\otimes 36}$ product state have identical single-site density matrices ($\rho_k = \mathbf{I}/2$ for every qubit $k$) and identical single-site energies $\langle H_k \rangle = 0$ for the clock Hamiltonian. The two-body correlators $\langle \sigma_j^z \sigma_k^z \rangle$ differ between cluster state and product state, but for nearest-neighbor Ising-type Hamiltonians, the energy difference is:

$$\Delta E_{AB} = \langle H_{\text{sys}} \rangle_A - \langle H_{\text{sys}} \rangle_B = \sum_{\langle jk \rangle} J_{jk} \underbrace{\left( \langle \sigma_j^+ \sigma_k^- \rangle_A - \langle \sigma_j^+ \sigma_k^- \rangle_B \right)}_{\text{same for both (both=0)}}. \tag{13}$$

For the clock qubit Hamiltonian with no $XX$ or $YY$ coupling (only $ZZ$ from the Rydberg interaction during gates, zero after gates), both states have the same mean energy after state preparation. This is the crucial property ensuring that the *mean-field* frequency shift is matched between branches.

## 3.2 Timing and observable

**Ramsey measurement protocol:**

1. Prepare Branch A (or B) state via the protocol above.
2. Apply global $\pi/2$ pulse on clock transition: $R_x(\pi/2)$.
3. Free precession for $T_{\text{Ramsey}} \in \{1, 10, 100\}$ ms.
4. Apply global $\pi/2$ pulse: $R_x(\pi/2)$ or $R_x(-\pi/2)$.
5. Measure population in $|1\rangle$ via site-resolved fluorescence.
6. Repeat $n_{\text{shots}} = 10^4$ times.

**Differential phase:**

$$\Delta\phi_{AB} = \phi^A - \phi^B = (\omega_{\text{eff}}^A - \omega_{\text{eff}}^B)\, T_{\text{Ramsey}}, \tag{14}$$

where $\omega_{\text{eff}}^{A,B}$ is the effective transition frequency in each branch.

**Signal formula:** The entanglement coupling predicts a frequency shift:

$$\frac{\delta\omega_{\text{hf}}}{\omega_{\text{hf}}} = \frac{\delta g_{00}}{2} = -\frac{\Phi_{\text{ent}}}{c^2}, \tag{15}$$

where $\Phi_{\text{ent}}$ is the effective gravitational redshift potential from the entanglement source. In the Newtonian limit, from Section 4:

$$\Phi_{\text{ent}} = -\frac{\kappa_{\text{eff}}\, M_{\text{ent}}}{R_{\text{eff}}}, \tag{16}$$

where $R_{\rm eff} \sim a = 5\,\mu$m is the characteristic distance scale within the array and $M_{\rm ent} = M^A_{\rm ent} - M^B_{\rm ent} = 6k_{\rm B}\ln 2$ (the differential entanglement). The measurable quantity translates to a bound:

$$\boxed{|\tilde{\kappa}_{\rm eff}|\,\frac{\Delta M_{\rm ent}}{R^2_{\rm eff}} < \frac{|\Delta\phi_{AB}|_{\rm sens}\,8\pi G k_{\rm B}\ln 2}{c^2\,\omega_{\rm hf}\,T_{\rm Ramsey}}\cdot\frac{1}{a}.} \tag{17}$$

Substituting: $|\Delta\phi|_{\rm sens} = 10^{-3}\,$rad (quantum projection noise at $N = 36$ qubits with $10^4$ shots), $T_{\rm Ramsey} = 0.1\,$s, $a = 5\times 10^{-6}\,$m, $\omega_{\rm hf} = 2\pi\times 6.835\times 10^9\,{\rm s}^{-1}$:

$$|\tilde{\kappa}_{\rm eff}|\,\frac{\Delta M_{\rm ent}}{a^2} < \frac{10^{-3}\times 8\pi\times 6.67\times 10^{-11}\times 9.57\times 10^{-24}}{(3\times 10^8)^2\times 2\pi\times 6.835\times 10^9\times 0.1}\times\frac{1}{5\times 10^{-6}} \approx 3\times 10^{-52}\ {\rm kg}^{-1}\,{\rm m\,s}^2. \tag{18}$$

### 3.3 Systematic error budget: fatal vs. non-fatal

**The two fatal systematics in detail:**

*Fatal Systematic 1 — Atom number mismatch.* If Branch A and Branch B have different mean atom numbers (e.g., due to different preparation-induced loss), then the density-density interaction shifts $(\delta\omega) \propto a^{(3)}_{\rm Rb}\langle n^2\rangle$ differ between branches, producing a spurious differential phase. Mitigation: post-select on runs where both branches have the same atom count (verified by site-resolved imaging). This reduces statistics by a factor of $\sim$ (fraction of runs with equal count), but does not reduce sensitivity per run.

*Fatal Systematic 2 — Unverified entanglement.* If the claimed cluster state is actually a product state (due to gate failures), the differential entanglement $\Delta M_{\rm ent} = 0$ and any measured phase shift would be misattributed to the coupling. Mitigation: run entanglement witness measurements before every Ramsey sequence. The witness operator $W_{\rm Cl} = \sum_k(\mathbf{I} - \sigma^z_k\sigma^z_{k+1})/2$ should satisfy $\langle W_{\rm Cl}\rangle < 0$ for any genuine cluster state. Runs failing this criterion are excluded.

**What a null result means.** If $\Delta\phi_{AB} = 0$ to within $|\Delta\phi|_{\rm sens} = 10^{-3}\,$rad after $10^4$ shots, this establishes the bound (18). This is a meaningful constraint on the observable combination $|\tilde{\kappa}_{\rm eff}|\,\Delta M_{\rm ent}/R^2_{\rm eff}$ regardless of the value of $\alpha_{\rm screen}$ or the Planck-scale renormalization of $\tilde{\kappa}$. The paper correctly frames this as its primary experimental output.

## 4 Task 5: Publication Strategy and Gap Analysis

### 4.1 Paper I: Theory Letter (4–6 pages)

**Target journal:** Physical Review Letters or Physical Review D (Letters format).

**Core claim:** A dimensionally consistent, covariantly conserved phenomenological extension of GR in which operationally defined internal entanglement entropy density contributes to the stress-energy tensor. The key structural results are: (a) definition of $S_{\rm ent}$ as internal inter-subsystem entanglement entropy density (not total or system-environment entropy); (b) the minimal effective action (1) with $\phi$ as an external source field; (c) $\Delta a \propto M_{\rm ent}/R^2$ for bounded sources; (d) the fifth-force prediction from Eq. (4); (e) the observable combination $|\tilde{\kappa}_{\rm eff}|\,\Delta M_{\rm ent}/R^2_{\rm eff}$ as the experimental falsification target.

**What to keep from the white paper:** Sections 2, 3 (with improvements from Task 2), 4, and 6.1–6.2.

**What to cut:** The Lindblad model details (move to Paper II). The macroscopic BEC discussion (keep only a brief note). The detailed experimental protocol (Paper III).

**Minimal additional results before submission:**

- Resolve the action tension (Task 2, Resolution A): one paragraph.
- Add the fifth-force observable: one boxed equation.
- State the metric signature explicitly.
- Verify the Bose et al. (2023) citation.

## 4.2 Paper II: Technical Backup (20–30 pages)

**Target journal:** Physical Review D (full article) or Classical and Quantum Gravity.

**Core content:**

- Full derivation of the effective action and covariant conservation (Section 3 expanded)
- Complete OP3 results (Stages 1–3 from Task 3): $\alpha_{\text{screen}}(N, \tau)$ for 50-qubit transmon cluster states
- OP8: Bianchi identity consequences and fifth-force bounds
- OP10: Planck-scale renormalization group power-counting (estimate whether $\tilde{\kappa}$ at the Planck scale renormalizes to an observable value at laboratory scales)
- Regime hierarchy table: $\alpha_{\text{screen}}$ for few-qubit (Regime I), mesoscopic (Regime II), macroscopic (Regime III)

**Minimal additional results before submission:**

- Stages 1–2 of the OP3 plan (Months 1–4): numerical Lindblad for $N \leq 20$.
- A power-counting estimate of OP10: write $\tilde{\kappa}_{\text{lab}} = \tilde{\kappa}_{\text{Planck}} \times (E_{\text{lab}}/E_P)^{\alpha}$ and estimate $\alpha$ from dimensional analysis.
- An explicit calculation of the fifth-force bound: for a test mass near a 50-qubit chip, what is the anomalous acceleration predicted by Eq. (4)?

## 4.3 Paper III: Experimental Proposal (12–15 pages)

**Target journal:** Physical Review A (experimental) or npj Quantum Information.

**Core content:**

- Full specification of the 36-qubit neutral-atom Ramsey experiment (Task 4)
- Explicit formula (17) relating measured $\Delta\phi_{AB}$ to the bound on $|\tilde{\kappa}_{\text{eff}}|\,\Delta M_{\text{ent}}/R_{\text{eff}}^2$

- Complete systematic error budget (Table 1)
- The OP3 prediction for $\alpha_{\text{screen}}$ (from Stage 3 of Task 3) as a theoretical input to the experiment
- A comparison section: what would it mean if $\alpha_{\text{screen}}^{\text{measured}} \neq \alpha_{\text{screen}}^{\text{Lindblad}}$?

**Minimal additional results before submission:**

- Stage 3 of the OP3 plan (MPO-Lindblad for $N = 50$): numerical prediction of $\alpha_{\text{screen}}(50, \tau)$.
- Explicit calculation of the mean-field energy matching for the cluster vs. product state pair (verify $\Delta E_{AB} = 0$ for the chosen Hamiltonian).
- An estimate of the quantum projection noise floor: $|\Delta\phi|_{\text{sens}} \sim 1/\sqrt{N_{\text{atoms}}\, n_{\text{shots}}}$.

### 4.4 Timeline

| Months | Deliverable | Paper |
|---|---|---|
| 0–1 | Action tension resolved; $S_{\text{ent}}$ defined; fifth-force equation added | Paper I |
| 1–2 | Paper I submitted | |
| 1–4 | OP3 Stage 1–2: Lindblad numerics $N \leq 20$ | Paper II |
| 4–8 | OP3 Stage 3: MPO-Lindblad $N = 50$; OP10 power-counting | Paper II |
| 4–6 | Mean-field energy matching; noise floor estimate | Paper III |
| 8 | Paper II and Paper III submitted | |

## Summary: The Three Most Important Actions

1. **Fix the action tension (one afternoon).** Change the effective action to treat $\phi = S_{\text{ent}}$ as an external source field (not varied), and note that conservation is guaranteed by $\nabla^\mu G_{\mu\nu} \equiv 0$, not by a $\phi$ equation of motion. Use the replacement text in Task 2. This removes a logical inconsistency and makes Paper I submittable.

2. **Specify the bipartition (one sentence).** The scalar field $S_{\text{ent}}(x)$ depends on the choice of bipartition $A|B$. State explicitly that the canonical choice for a 2D array is a geometric half-plane cut, and that the results are insensitive to the exact cut location for translationally invariant systems. This makes $S_{\text{ent}}$ well-defined as a spacetime scalar field.

3. **Run Stage 1 of OP3 (two months).** Solve the Lindblad master equation numerically for $N = 10$ transmon qubits in a cluster state and compute $\alpha_{\text{screen}}(10, T_\phi)$. This single number determines whether the experimental program is viable at NISQ scales and whether Papers II and III are worth pursuing. If $\alpha_{\text{screen}}(10, T_\phi) \lesssim 10^{-6}$, the framework's near-term experimental program requires fundamental revision. If $\alpha_{\text{screen}} \gtrsim 10^{-2}$, the path forward is clear.

# References


[1] T. Jacobson, *Phys. Rev. Lett.* **75**, 1260 (1995).

[2] E. Verlinde, *SciPost Phys.* **2**, 016 (2017).

[3] P. Asenbaum *et al.*, *Phys. Rev. Lett.* **125**, 191101 (2020).

[4] P. T. Touboul *et al.*, *Phys. Rev. Lett.* **129**, 121102 (2022).

[5] R. Blume-Kohout *et al.*, *Nat. Commun.* **8**, 14485 (2017).

[6] J. Haah *et al.*, *IEEE Trans. Inf. Theory* **63**, 5628 (2017).

[7] J. Hauschild and F. Pollmann, "Efficient numerical simulations with Tensor Networks: Tensor Network Python (TeNPy)," *SciPost Phys. Lect. Notes* **5** (2018).

[8] M. Saffman, *J. Phys. B* **49**, 202001 (2016).

[9] G. Vidal, *Phys. Rev. Lett.* **93**, 040502 (2004).

[10] J. Haegeman *et al.*, *Phys. Rev. B* **94**, 165116 (2016). [TDVP for MPOs]

Table 1: Systematic error budget for the 36-qubit neutral-atom Ramsey experiment.

| Systematic | Origin | Mitigation | Fatal? |
|---|---|---|---|
| Differential Zeeman shift | Magnetic field $B(t)$ fluctuations | Clock transition ($m_F = 0$) insensitive to first order; active shielding to $\delta B < 1\,\mathrm{nG}$ | **No** |
| Differential AC Stark (clock laser) | Laser intensity $I(t)$ fluctuations | Common-mode rejection: same laser for A and B; interleave shots | **No** |
| Differential two-body correlator frequency shift | $\langle \sigma_j^z \sigma_k^z \rangle_A \neq \langle \sigma_j^z \sigma_k^z \rangle_B$ during Ramsey | Choose A,B states with matched $\langle \sigma_j^z \sigma_k^z \rangle$ (cluster vs. product both give zero off-resonant Ising coupling in free precession) | **No** (for chosen state pair) |
| Rydberg admixture during free precession | Rydberg laser not fully off | Verify zero admixture via spectroscopy; separate Rydberg pulse from Ramsey | **No** |
| Differential SPAM errors | State-dependent loss during readout | Symmetrize A/B preparation schedule; verify SPAM via reference states | **No** |
| Differential mean-field shift from density-density interactions | $n_{\mathrm{Rb}}^A \neq n_{\mathrm{Rb}}^B$ | Same trap configuration, same atom number; post-select on identical atom count | **Yes** (if different $N_{\mathrm{atoms}}$) |
| Motional state difference (entangled vs. product) | Different mean kinetic energy for A vs. B | Verify temperature equivalence via sideband spectroscopy; Doppler-free clock transition | **No** |
| Entanglement verification failure | SPAM + readout errors prevent confirming entanglement depth | Use entanglement witnesses with SPAM correction; require $W < 0$ with $> 5\sigma$ confidence before including run | **Yes** (if unverified) |